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ORIGINAL ARTICLE

A note on weak shock wave reflection

D. P. Viero · F. M. Susin · A. Defina

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Abstract This work discusses the possibility of reconstructing, both numerically and experimentally, the steady state flow field and shock reflection pattern close to the triple point of von Neumann, Guderley and Vasilev reflections. First, a criterion for the orientation of shock wave fronts, even in the case of subcritical/subsonic flow downstream the front, is introduced and formalized. Then, a technique for obtaining a close view of the above reflection patterns centered about the triple point is described and a numerical example, within the framework of shallow water flow, is presented and discussed.

Keywords Shock wave orientation · Weak shock reflection · Von Neumann paradox · Supercritical open channel flow

1 Introduction

Since the work of Colella and Henderson [1], which inspired much of the recent research on weak shock wave reflection off a wedge, a number of important contributions have appeared in the literature [2–9]. We can now claim that for weak shock waves and small wedge angles (i.e., under the von Neumann paradox conditions) the solution to the reflection patterns close to the triple point is provided by the four-wave theory, originally proposed by [10], and markedly improved and formalized by [9].

The basic structure of the reflection pattern is shown in Fig. 1a: the incident shock (I) reflects at the triple point T and the reflected shock (R) has a direction such that the flow behind it (region 2) is uniformly critical (sonic, in gas flow) close to the triple point. Region (2) is bounded downstream by an expansion fan (E) originating at the triple point with the leading characteristic normal to the flow direction in region (2). A slip stream (S) also originates at the triple point, and separates the flow of region (3) from the flow of region (4) with a lower Froude number (the Froude number is defined as the ratio of flow velocity to the gravitational wave velocity; it is equivalent to the Mach number, in gas flow). Finally, a shock wave of the same family as I and R, separates the incoming flow (region 0) from the flow of region (4). This shock wave is referred to as Guderley stem (G) if the flow below the slipstream is supercritical (Fig. 1b), otherwise it is called Mach stem (M) (Fig. 1c).

It is worth pointing out that the four-wave theory only provides a solution for the weak shock wave reflection problem close to the triple point; the solution to the full reflection problem requires additional items such as a diffused compression wave through the supercritical patch which hits and gently turns the Guderley stem [11] or the existence of a sequence of triple points and supercritical patches, all attached to the Guderley stem [7,8,11,12]. In the latter case the reflection pattern is termed Guderley Mach reflection [12,13].

The different flow state below the slip stream and close to the triple point (i.e., supercritical or subcritical) also discriminates between two different reflection patterns which are referred to as Guderley and Vasilev reflection, respectively [14].

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D. P. Viero (✉) · F. M. Susin · A. Defina
Department ICEA, University of Padova,
via Loredan 20, 35131 Padova, Italy
e-mail: daniele.viero@unipd.it

F. M. Susin
e-mail: francescamaria.susin@dicea.unipd.it

A. Defina
e-mail: andrea.defina@unipd.it

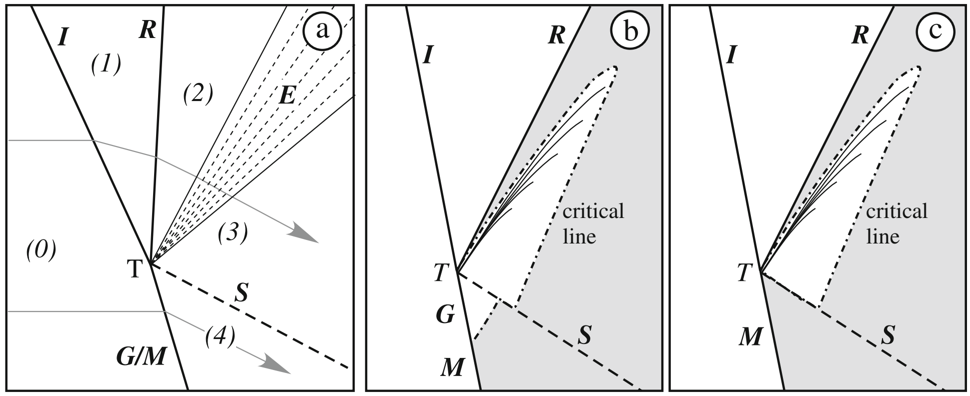

Fig. 1 Reflection patterns of weak shock waves close to the triple point: **a** schematic illustration of the four-wave configuration; **b** Guderley reflection; **c** Vasilev reflection. *Critical line* is equivalent to *sonic line* in gas flow. *The shaded area* has $F < 1$ (Froude number F is equivalent to Mach number in gas flow). Flow is from *left to right*

Although considerable progress has been made in characterizing these complex reflection patterns, a number of open issues still remain, e.g., the reason for the Guderley stem being curved, the possibility of having an infinite sequence of triple points and patches, the nature of interactions between weak shocks, expansion wave and subcritical/subsonic flow, the impact of viscosity and real fluid effects.

Recent theoretical (e.g., [15]) and experimental [16] investigations confirm the great interest in the details of the weak shock wave reflection close to the triple point.

The main problem facing a detailed study of these reflection patterns is given by the size of the supercritical patch which, even at small Froude/Mach number, is extremely tiny [11,17].

Therefore, studying the details of these reflection patterns, and the flow field in the neighborhood of the triple point and within the supercritical patch, is quite a demanding task both in numerical and experimental approach.

In this work we suggest a technique for reconstructing the shock wave pattern and the flow field close to the triple point of weak shock wave reflection. This technique is based on a suitable formalization of the 'shock wave front orientation', which is preliminarily introduced. We then present and discuss a numerical example within the framework of shallow water flow.

2 Orienting shock wave front

The idea of orienting shock wave front is somehow embedded or intrinsically contained in any discussion on shock waves: when we label a front as "incident" or "reflected" we implicitly claim its orientation as being toward or away from the reflecting wall, respectively. The issue of defining shock wave orientation is however not trivial when the flow downstream a shock is subcritical (i.e., subsonic in gas flow).

If the flow on both sides is supercritical, every disturbance must be propagated along the shock in the direction of the tangential component of the fluid velocity. Accordingly,

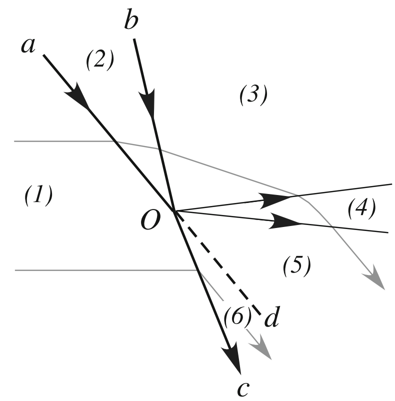

Fig. 2 Collisions of two shock waves (*Oa* and *Ob*) resulting in one expansion fan (region 4) and one shock wave (*Oc*). Also, a slip stream (*Od*) forms between the expansion fan and the shock wave *Oc*. Flow is from *left to right*, and *grey lines* denote streamlines. (After Landau and Lifshitz [18])

Landau and Lifshitz [18] define the shock front direction to be the direction of propagation of disturbances.

When the downstream flow is subcritical, however, disturbances are propagated in both directions along the shock wave and there is strictly no meaning in speaking of its "direction" within the above definition framework. Nevertheless, in the aim of their discussion on front intersection, Landau and Lifshitz [18, §110] extend the definition of shock wave orientation to include the case of downstream subcritical flow by assuming that shock waves, in this case, "leave the intersection" since disturbances can be propagated along the front away from the intersection point.

The latter is a useful practical definition which, however, seems not to be generally applicable. Among the many examples of front intersection discussed by Landau and Lifshitz [18], the one they show in their Figure 101*b* (see Fig. 2) deserves some discussion. The example considers the collision of two shocks (*Oa* and *Ob*) which results in one shock wave (*Oc*) and one rarefaction wave (region 4). Also, a tangential discontinuity (the slip stream *Od*) must be formed, and it must lie between the expansion fan and the shock wave *Oc*.

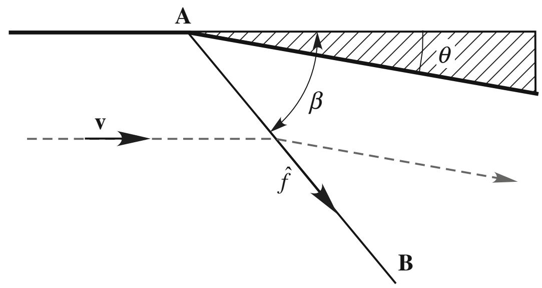

Fig. 3 Schematic illustration of a shock wave front off a wedge, with notation

It is worth noting that the flow in regions (3) and (6) can be subcritical (an example is given in Sect. 3): in this case, the "direction" of the shock wave Ob separating flow regions (2) and (3), does not match the above criterion, since the shock wave is "approaching" and not "leaving" the intersection point O, as denoted by the arrows in Fig. 2.

Henderson and Menikoff [19] suggest, in accordance with Landau and Lifshitz [18], that shock waves can be classified as incoming or outgoing according to the direction in which information flows. However, they assume that information propagates with flow velocity rather then, as suggested in [18], with the velocity of propagation of disturbances and say that "a wave is incoming if the component of the velocity in the direction of the front points towards the node (i.e., the intersection point), otherwise the wave is outgoing".

The lack of a reliable criterion defining the shock front orientation, also in presence of subcritical flows, sometimes leads to inconsistent statements such as "the front is sloping forwards in the reference frame of the upstream velocity". We show that this apparent forwards sloping results from the inappropriate assumed front orientation. Importantly, we show that the proposed orientation criterion inherently suggests a way to easily reconstruct weak shock wave reflection patterns, either with numerical models or in laboratory experiments, in order to improve our understanding of this issue.

In this work we release the idea of relating shock wave orientation to the direction in which perturbations must be propagated, and approach the problem from a different point of view.

When we say that a shock wave originates at the leading edge A of a compressive wedge (see Fig. 3), we implicitly suggest the idea that the front direction points away from the wedge A, i.e. that the front is oriented toward point B, no matter if the flow behind the shock wave is supercritical or subcritical. This intrinsic orientation results from that the shock front AB can be produced in A, e.g. by a deflecting wall, but cannot be produced in B anyway.

We then define the shock wave "orientation", i.e., the sense of the shock front, assuming that a shock front is oriented from the end point where it can be triggered (e.g. by a deflecting wall) to the opposite end point, no matter if the flow behind it is supercritical or subcritical.

In steady flow a shock wave originates where the flow is deviated by an angle θ such that $0 \leq \theta \leq \theta_D$, θ_D being the detachment angle. In the frame reference of upstream flow direction, a shock wave front is inclined at an angle β (see Fig. 3) in the range: $\arcsin(1/F_0) \leq \beta \leq \pi/2$, F_0 being the upstream Froude number. Therefore, a shock wave is always inclined in the flow direction.

The proposed criterion can then be expressed by the condition (Fig. 3)

$$\hat{f} \cdot \mathbf{v} > 0 \tag{1}$$

where $\hat{f}$ is the shock front unit vector and $\mathbf{v}$ is the upstream velocity. Shock front orientation is indeterminate when the above dot product is zero. Therefore, a straight normal shock cannot be oriented.

Given a finite length shock wave, such that the constraint (1) is verified all along the front, front orientation allows to establish at which end point the shock wave originates.

If the upstream flow is uniform and the downstream flow is supercritical, every disturbance must be propagated along the shock wave in the direction of the tangential component of the fluid velocity and the shock front is straight. Otherwise, as disturbances propagating in the downstream subcritical flow can move in any direction along the front, the shock itself is curved.

It is possible, for a curved shock, that the above constraint (1) is not verified at one point along the shock wave front where $\hat{f} \cdot \mathbf{v} = 0$. An example is given by the reflected shock in the von Neumann reflection (Fig. 4a). In this reflection pattern a point V along the reflected shock exists such that the reflected shock is a normal shock in V (i.e., $\hat{f} \cdot \mathbf{v} = 0$). We refer to V as the vertex of the reflected shock.

If we divide the flow field into two parts, one at the left and one at the right of the streamline through point V (Fig. 4a), we see that the left part contains the only branch Vb of the reflected shock. This branch can be triggered at the end point V, therefore it is oriented from V to b. In the right part of the flow field we have a shock wave pattern produced by the collision of shock waves aT and VT. The latter, i.e., the inner branch of the reflected shock, again can be triggered at the endpoint V and it is thus oriented from V to T. In this sense, the vertex V can be thought of as the origin of the two branches of the reflected shock: an inner branch oriented toward the triple point T, and an outer branch having the opposite direction.

Therefore, the inner branch is not strictly a reflected shock but it is rather a shock originating at V and colliding the incident shock wave at the triple point.

Other examples are shown in Fig. 4b, c. The Guderley reflection (Fig. 4b) is very similar to the von Neumann reflection; the main difference stems from the presence of

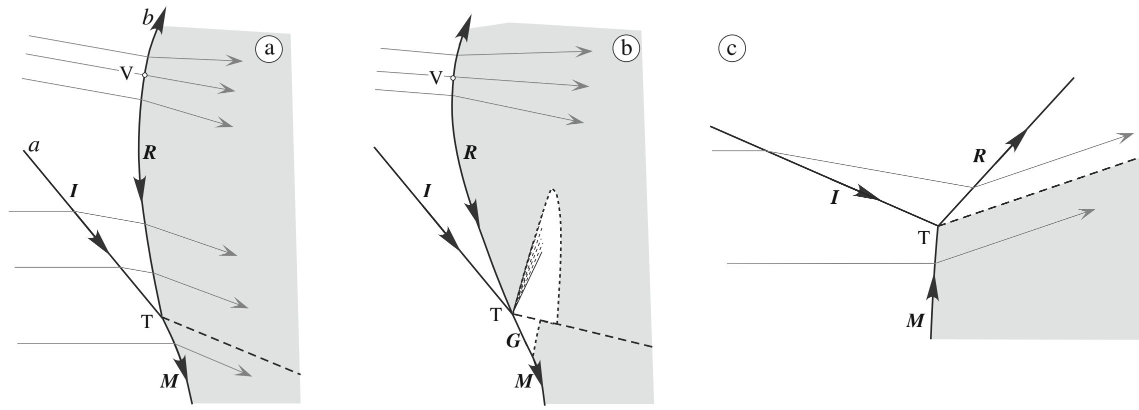

Fig. 4 Flow field and shock wave pattern close to the triple point T of (**a**) von Neumann reflection, (**b**) Guderley reflection, and (**c**) inverse reflection. *I* and *R* are the incident and reflected shock, *M* is the Mach stem, *S* the slipstream, *G* the Guderley stem. *The shaded area* denotes subcritical (subsonic) flow

a supercritical patch close to the triple point and attached to the Mach stem. It is worth noting that the orientation of the reflected shock as given by constraint (1) is consistent with the theory stating that two colliding shocks of the same family (i.e., the incident and the inner branch of the reflected shock) can result in a rarefaction wave of the opposite family (i.e., the expansion fan centered at the triple point) in addition to a stronger shock of the original family (i.e., the Guderley or Mach stem), see e.g., [18,21] and Fig. 2b. In the inverse irregular reflection, which can be found in the reflection of asymmetric shock waves in steady flows [22,23], the Mach stem is oriented toward the triple point (Fig. 4c).

The overall picture agrees with that described by Semenov et al. [20] who state that these reflection configurations "contain an *incoming* shock wave as a reflected one".

3 Modelling weak shock wave reflection pattern

As we stated in Sect. 1, experimental and numerical modeling of the details of weak shock wave reflection patterns and flow field in the neighborhood of the triple point, is quite a demanding task. On the contrary, we show that within the framework of oriented shock fronts it is possible to construct the detailed flow field and shock wave pattern close to the triple point of Guderley, Vasilev, and von Neumann reflection with minor computational effort.

This can be accomplished because both the incident and the inner branch of the reflected shock are oriented toward the triple point and the Mach (or Guderley) stem is oriented away from the triple point. As such, the shock pattern configuration looks more like a front intersection problem rather than a reflection problem. An example of Guderley reflection reconstruction within the framework of shallow water flow is given here.

The flow across an oblique shock wave is governed by the Rankine–Hugoniot jump relations which, for a shallow free surface flow, read [11],

$$\frac{h}{h_0} = \frac{1}{2}\left[-1 + \sqrt{1 + 8F_0^2 \sin^2(\beta)}\right] \tag{2}$$

$$\tan(\theta) = \tan(\beta)\frac{\sqrt{1 + 8F_0^2 \sin^2(\beta)} - 3}{2\tan^2(\theta) - 1 + \sqrt{1 + 8F_0^2 \sin^2(\beta)}} \tag{3}$$

$$F = F_0 \frac{\cos(\beta)}{\cos(\beta - \theta)}\sqrt{\frac{\tan(\beta - \theta))}{\tan(\beta)}} \tag{4}$$

where h is the flow depth, F is the Froude number, and the subscripts '0' refers to the flow upstream of the shock. In the limit of vanishingly small shocks, the above equations reduce to

$$\theta = \sqrt{3}\arctan\left(\frac{\sqrt{F^2 - 1}}{\sqrt{3}}\right) - \arctan\left(\sqrt{F^2 - 1}\right) + \text{const} \tag{5}$$

$$\beta = \arcsin\left(\frac{1}{F}\right) \tag{6}$$

$$h = \text{const} \cdot \frac{2}{2 + F^2} \tag{7}$$

On combining (2) and (3) to eliminate β the shock polar equation is obtained, which relates the relative water depth h/h_0 to flow direction θ, for any given Froude number of the incoming flow.

Figure 5a gives an example of the graphical solution to the four-wave theory for a Guderley reflection, close to the triple point. The solution uses the incident (I) and reflected (R) shock polar curves, and the curve for the expansion fan (E) starting from the incident critical condition c_R. This curve is obtained by combining (5) and (7) to eliminate F. For details, the reader is referred to [9,11], and [14].

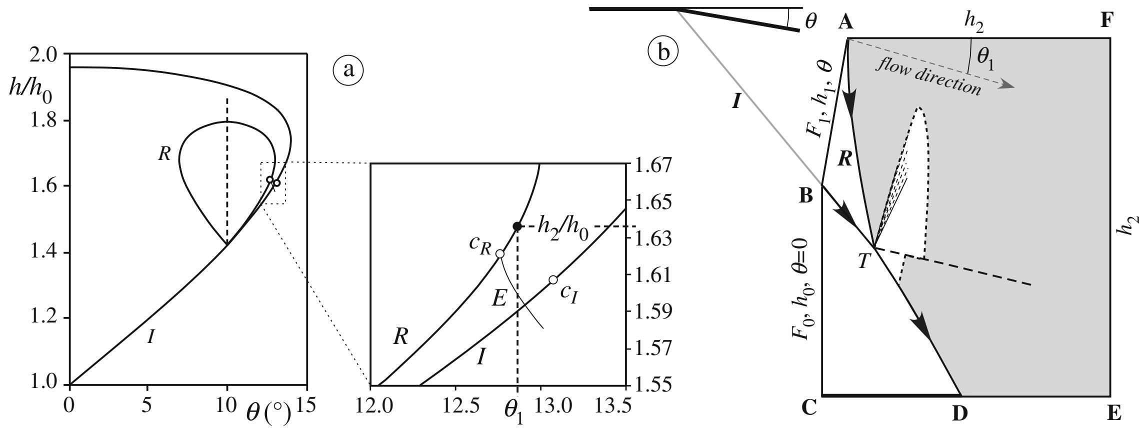

Fig. 5 **a** Shock polar representation; c_R and c_I are critical conditions along the reflected and incident shock polars, respectively. **b** Schematic illustration of the computational domain, shock wave pattern, and prescribed boundary conditions. Flow is from *left to right*. *Shaded area* denotes subcritical flow condition

Starting from the solution provided by the four-wave theory we construct a first guess of the shock wave pattern as follows.

We construct the domain labelled $ABCDEF$ in Fig. 5b, with a leading deviation θ triggering the incident shock front I outside of the domain. Line AB is taken to be normal to the flow direction behind the incident shock, i.e., normal to the deflecting wall.

To produce the inner branch of the reflected shock (R), the flow along AF and close to A must have the direction $\theta_1 > \theta$ (Fig. 5b). Based on the behavior of the solution with increasing of the distance from the triple point as discussed by [17] (see their Fig. 6), we arbitrarily choose the deviation θ_1 to be slightly greater than the deviation corresponding to the critical condition c_R of Fig. 5a. The flow direction θ_1 is then forced by imposing along the boundary AF the corresponding water depth h_2 as shown in Fig. 5a.

The position of point B is such that the incident shock and the inner branch of the reflected shock meet at the triple point T located approximately midway between the upper and lower boundary of the domain.

Point D is then obtained from the intersection of the Mach stem with the boundary CE. The Mach stem is oriented away from the triple point and toward the lower boundary of the domain. The position of point D is thus specified not only by the position of the triple point (which cannot be determined exactly because the inner branch of the reflected shock R is slightly curved) but also by the direction and shape of the Mach stem. Therefore, the position of point D cannot be accurately predicted *a priori* and it is determined by trial and error as discussed later in the text.

The left boundary BC is a supercritical inflow where all variables (h_0, F_0, and $\theta = 0$) are specified. Along the upstream boundary AB, we prescribe water depth h_1, Froude number F_1, and flow direction θ, as provided for the flow field behind the incident shock wave by the solution of (2)–(4).

Along the boundary $AFED$ only water depth h_2 is prescribed (boundary condition for velocity is here redundant). This boundary is supposed to be far enough from the triple point so that the arbitrarily prescribed subcritical flow had a minor impact on the solution close to the triple point.

Finally, along the boundary CD the inviscid (i.e., free slip) solid wall condition is imposed.

3.1 The numerical model

The unsteady depth-averaged shallow water equations (SWE) are used to solve for the shock reflection patterns in a supercritical channel flow within the domain shown in Fig. 5b. The conservative form of depth-averaged SWE are written in vector form as

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \mathbf{F} = \mathbf{S} \tag{8}$$

where

$$\mathbf{U} = \begin{pmatrix} h \\ uh \\ vh \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} hu & hv \\ hu^2 + \frac{1}{2}gh^2 & huv \\ huv & hv^2 + \frac{1}{2}gh^2 \end{pmatrix} \tag{9}$$

and u, v are the flow velocity components in x, y directions, respectively, h is the water depth and g is gravity. On neglecting viscosity and assuming an horizontal bottom, the source term S in (9) is identically zero.

The SWE are solved with a second-order accurate Godunov-type finite volume model on unstructured triangular grid composed of approximately 8×10^4 nodes and 1.6×10^5 elements. Further details are given by [11].

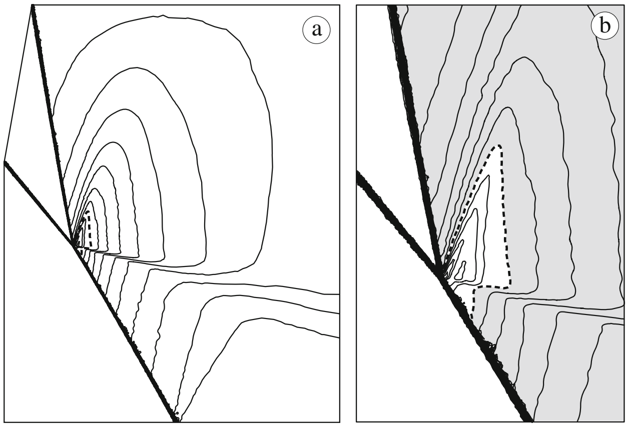

Fig. 6 **a** Reconstructed Guderley reflection pattern for $F_0 = 1.7$ and $\theta = 10°$. The *solid lines* are iso-Froude contours (contour spacing is 0.005); *the dashed line* is the critical line. **b** Enlarged view close to the triple point; shaded area has $F < 1$

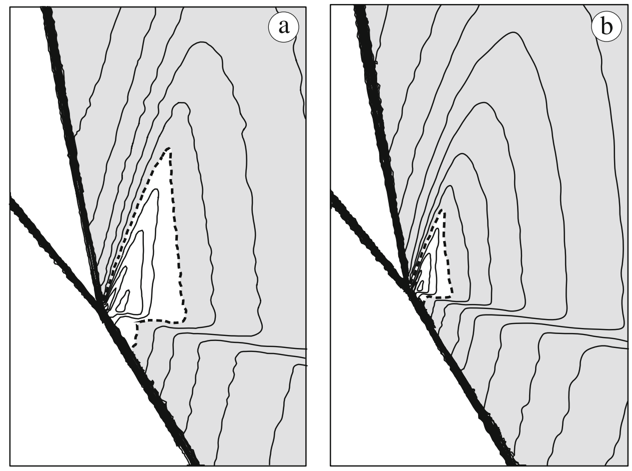

Fig. 7 Close view of the reconstructed Guderley reflection pattern for $F_0 = 1.7$ and $\theta = 10°$. *The solid lines* are iso-Froude contours (contour spacing is 0.005); *the dashed line* is the critical line. **a** Solution for the case of enlarged domain downstream of the reflected shock and the Mach stem. **b** Solution for the case of downstream imposed water depth $h_2/h_0 = 1.67$

3.2 The computations

We reconstruct the Guderley reflection pattern in a supercritical open channel flow with $F_0 = 1.7$ and $\theta = 10°$. From (2)–(4) we have $F_1 = 1.193$ and $h_1/h_0 = 1.42$ which are used as upstream boundary conditions along the boundary AB of Fig. 5b. Critical point on the reflected shock polar (Fig. 5a) is $c_R = (12.767°, 1.621)$, then we arbitrarily choose $h_2/h_0 = 1.66$ which is slightly greater than 1.621.

Numerical results are shown in Fig. 6: Fig. 6a shows iso-Froude contours and the critical line, Fig. 6b is an enlarged view of the solution near the triple point. The computed supercritical patch has a shape that looks remarkably similar to the one discussed by [11] who resolved the same reflection problem. The structure of the flow field and the shock wave pattern compare favorably with their numerical results. Also, results are not affected if grid resolution is increased: only more sharp wave fronts can be observed.

We then perform checks to determine the sensitivity of the solution to the placement and size of the numerical boundary $DEFA$. In this study, we change the shape of the boundary $DEFA$ and greatly enlarge the area with subcritical flow behind the reflected shock and the Mach stem. Along the new downstream boundary we again impose $h_2/h_0 = 1.66$.

The results of this simulation are shown in Fig. 7a where iso-Froude contours are plotted over the same sub-domain as in Fig. 6b. A qualitative comparison between Fig. 6b and Fig. 7a shows the good agreement between the solutions. However, the size of the supercritical patch in Fig. 7a is slightly greater than that of Fig. 6b, thus suggesting that the extent of the computational domain downstream of the reflected shock wave and Mach stem is relatively small when compared to the size of the supercritical patch.

We then check the impact of the arbitrarily prescribed downstream water depth. We performed a simulation after increasing the imposed water depth to $h_2/h_0 = 1.67$. Figure 7b shows that increasing the water depth slightly moves the triple point leftward and largely reduces the size of the supercritical patch. However, the flow field and shock wave pattern look closely like those in Fig. 7a and, importantly, in this case the solution is found to be unaffected by the shape and size of the downstream subcritical domain.

Some discussion is finally devoted to the problem of determining the position of point D. As stated in Sect. 3, the position of point D cannot be predicted a priori because both the shock wave R and the Mach stem are curved shock fronts. The position of point D is determined by trial and error. As a first approximation its position is estimated by assuming that both the shock wave R and the Mach stem are straight shock waves. We then adjust the position of point D depending on the results of the simulation. If we locate point D downstream from its correct position then the Mach stem reflects off the impermeable wall CD and the reflected shock wave, which is unstable, migrates upstream cancelling the shock wave pattern by establishing subcritical flow all over the domain (see Fig. 8a). Otherwise, if we locate point D upstream from its correct position, at point D a shock wave emanates which interacts with the Mach stem and locally disturbs the flow field (see Fig. 8b). However, in this case, small errors in the position of point D have a minor impact on the flow configuration close to the triple point.

4 Conclusions

In this paper we have suggested a criterion to define the "orientation" of shock waves which allows to include the case of subcritical (subsonic) flow behind the shock front.

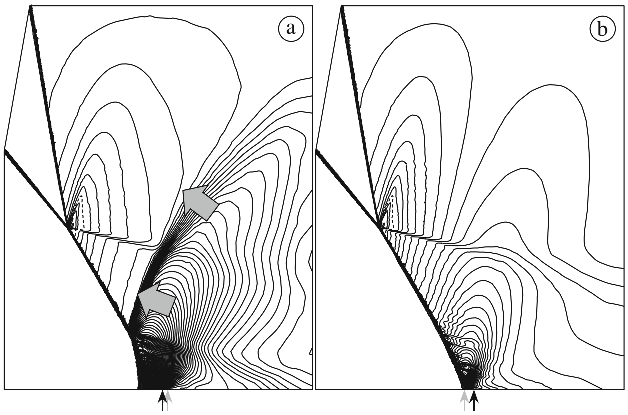

Fig. 8 Reconstructed Guderley reflection pattern for $F_0 = 1.7$ and $\theta = 10°$. *The solid lines* are iso-Froude contours (contour spacing is 0.005) and *the dashed line* is the critical line. *The vertical arrows* pointing the lower boundary of the computational domain denote the simulated (*gray arrow*) and correct (*black arrow*) position of point *D*. **a** Point *D* is located downstream from its correct position: the reflected shock wave generated by the collision of the Mach stem against the lower impermeable wall is unstable and migrates upstream as denoted by the the gray arrows. **b** Point *D* is located downstream from its correct position: a shock wave emanates from point *D* and after interacting with the Mach stem, diffuses into the subcritical flow downstream the reflected shock and Mach stem; in this case disturbances remain close to point *D* and do not affect the reflection pattern close to the triple point

Within the suggested framework of oriented shock wave, we have proposed a technique to reconstruct the steady shock wave pattern and flow field close to the triple point of weak shock wave reflection. The technique proved very effective and can be profitably used both in numerical and experimental investigations to explore further the nature of the flow field within and around the supercritical patch and to assess the impact of viscosity and real fluid properties.

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